

Questions on Lecture of cell structure

Choose the correct answer:

1. In eukaryotic cell DNA is present in:

a) In cytoplasm. b) In mitochondria. c) **nucleus** d) None of the above

2. In prokaryotic cell DNA is present in:

a) nucleoid region. b) mitochondria. c) nucleus

3. Water constitutes % of the body weight:

a) About 90% b) About 20% c) About 55% d) About 30%

4. Water constitutes % of the body weight in men:

a) About 90% b) About 20% c) About 60% d) About 30%

5. Water daily intake for adults should be not less than :

a) 10 liters b) 5 liters c) **At least 1.5 liters** d) 500 ml of water

6. A 70 kilogram man has about 45 liters of water in his body, 30 liters of them are present in

a- Intracellular Fluid b- Extracellular Fluid c- Interstitial Fluid

7. The organelle responsible for packaging in eukaryotic cells is :

a) Lysosome

c)Endoplasmic reticulum **d)Mitochondria**

8- The organelle responsible for ATP production in eukaryotic cells is :

a) Lysosome

c) Endoplasmic reticulum d) Mitochondria

9- The organelle responsible for protein synthesis in eukaryotic cells is :

a) Lysosome

c) **Endoplasmic reticulum**

10-is the main type of lipid in cell membrane.

a) phospholipids b) Glycolipids c) lipoproteins d) Glycerol

11- The hormone that maintain water at equilibrium in the body is:

a) ADH b) thyroid hormone c) insulin hormone

Compare between Prokaryotic and Eukaryotic cells

Feature	Prokaryotic Cells	Eukaryotic Cells
Genetic Material	Circular DNA, no nucleus	Linear DNA, nucleus
DNA Packaging	No histones	Histone-associated chromatin
Cell Membrane	Phospholipids only	Phospholipids + sterols
Organelles	Absent	Present
Ribosomes	70S	80S

Compare between Prokaryotic and Eukaryotic cells regarding genetic material

Prokaryotic cells	Eukaryotic cells
Circular DNA	Linear DNA
No histones	DNA wrapped around histones
Plasmid	No Plasmid

Match from column A to the suitable answer in column B

A	B
1- Mitochondria (d)	a- packaging
2- Endoplasmic reticulum (c)	b- digestion
3- Golgi apparatus (a)	c- protein synthesis
4- Lysosomes (b)	d- ATP synthesis

Define the following terms:

Apoptosis: It is a programmed cell death. It is a controlled process of cellular self-destruction

Neoplasia: is an abnormal accumulation of cells that occurs because of an imbalance between cellular proliferation and cellular attrition.

Complete the following space with suitable word:

- 1- It is a programmed cell death. It is a controlled process of cellular self-destruction... **Apoptosis**
- 2- is an abnormal accumulation of cells that occurs because of an imbalance between cellular proliferation and cellular attrition...
Neoplasia:.....
- 3- Thin envelopes that define the volume of cells and cellular organells.....**Membranes**.....

Enumerate 3 functions for the Membranes:

- 1-Membranes constitute the boundaries of cells and intracellular organelles.
- 2-Membranes have proteins that mediate and regulate the transport of metabolites, macromolecules and ions.
- 3- Exocytosis and endocytosis

.....

Questions on Lecture of carbohydrate biochemistry

Choose the correct answer:

1-Which one of the following sugars is a pentose.

a-Ribulose. b- Lactose c-Mannose. d-Glucose

2-Which one of the following sugars is a pentose.

a-Ribose.b- Lactose c-Fructose. d-Glyceralhyde

3-Which one of the following sugars is found in RNA?

a-Xylose. b-Maltose **c-Ribose** d- Glucose

4-Which one of the following sugars is found in DNA?

a- Deoxy-ribose. b- ribose c- ribulose d-xylulose

5- Which one of the following sugars is a monosaccharide?

6-a- Sucrose. **b- Glucose** c- Lactose d-Maltose

7-Which one of the following sugars is a monosaccharide?

a- Sucrose. **b-Fructose** c- Lactose d-Maltose

8-Which one of the following sugars is a disaccharide?

a- Starch. **b-Lactose** c- Galactose d-Glucose

9-Which one of the following sugars is a disaccharide?

a- Glycogen. **b-Sucrose** c- Fructose d-Glucose

10- Which one of the following sugars is a disaccharide?

a- Starch. **b-Maltose** c- Ribose d-Glucose

11- Which one of the following is homopolysaccharides:

a- Heparin **b-Glycogen**
c- proteoglycan d-Hyaluronic acid

12- Which one of the followings is homopolysaccharide:

a-Keratansulphate **b- Starch**

c-Chondroitinsulphate

d-Hyaluronic acid

13- Alpha 1-4 and alpha 1-6 glycosidic bonds are present in one of the following polysaccharides:

a- Amylose

b- **Glycogen**

c- Inulin

d-Cellulose

14- Alpha 1-4 and alpha 1-6 glycosidic bonds are present in one of the following polysaccharides:

a- Amylose

b- **Starch**

c- Hyaluronic acid

d-Cellulose

15- Which of the following is a micronutrient?

a.Carbohydrates

b. Fat

c. Protein

d.Vitamins and minerals

16- Dietary carbohydrates yield..... energy/gram

a.4Kcal

b.7Kcal

c.9Kcal

d.12Kcal

17- Dietary lipids yield energy/gram

a.4Kcal

b.7Kcal

c.**9Kcal**

d.12Kcal

18- Dietary proteins yield energy/gram

a.4Kcal

b.7Kcal

c.9Kcal

d.12Kcal

19- The non reducing disaccharides is:

a)**Sucrose**

b) lactose

c)fructose

d) Glucose

20- The most preferable source of energy for brain is:

a) **Glucose**

b) lipids

c) fructose

d) proteins

21- The only source of energy for RBCs is:

a) **Glucose**

b) lipids

c) fructose

d) protiens

22- Which of the following has anticoagulant properties:

- a) Heparin b) Hyaluronic acid c) Chondroitin sulfates

23- Carbohydrates that are formed of (3-10) sugar units linked by glycosidic linkage are called:

- a- Monosaccharides b- Oligosaccharides
c- Disaccharides d- Polysaccharides

24- Which one of the following is considered ketose?

- a- Glucose b- Glyceraldehyde
c- Fructose d- Erythrose

25- monosaccharides linked together by.....to form disaccharides and polysaccharides

- a) Ester bond b) glycosidic bond c) peptide bond

Complete the following space with suitable word:

1-Compounds that are needed in amounts of a gram or more per day Macro nutrients... For example, carbohydrate, protein, fat.

2- Compounds that are needed in amounts of a milligram or less per day Micro nutrients... For example, vitamins, minerals,

3- Two examples for hexoses include ...glucose, fructose.....

4- Examples for pentoses include,,,,ribose, deoxyribose

.....

Enumerate functions for the following:

A) Function of carbohydrates:

1- Glucose is the most important dietary carbohydrate and the major

metabolic fuel to the body.

2-Glycogen is a storage form of carbohydrates in liver and muscles

3-Ribose and deoxyribose are important for synthesis of nucleic acids (RNA&DNA)

4-Galactose is important for synthesis of lactose of milk.

5-Fructose is the fuel sugar for sperms

B) Function of pentoses (ribose and deoxyribose)

1- pentoses enters in the structure of DNA (deoxyribose) and RNA (ribose)

2- pentoses enters in the structure of important free nucleotides such as ATP, GTP and coenzymes NAD , NADP and FAD, FMN.

C) Function of Glycoproteins:

- Structural function e.g. cell walls, collagen, elastin, fibrin and bone matrix.
- Lubricant and protective functions e.g. mucins and mucous secretions.
- Immunologic function e.g. immunoglobulin.
- Hormonal function e.g. thyroid stimulating hormone (TSH).

Match from column A to the suitable answer in column B

A		B
1- Glucose	(d)	a- Anticoagulant
2- Fructose	(c)	b- enter in structure of RNA
3- Heparin	(a)	c- Important energy source for sperms
4- Ribose	(b)	d- Important energy source for brain
5- Lactose	(f)	e- a storage form of carbohydrates in liver and muscles
6- Gycogen	(e)	f- milk sugar

.....

Questions on Lecture of lipid biochemistry

Choose the correct answer:

1- Essential Fatty acid that should be in human diet include one of the following:

- a) Palmietic acid b) stearic acid c) oleic acid d) linoleic acid

2- Essential Fatty acid that should be in human diet include one of the following:

- a) Palmietic acid b) stearic acid c) oleic acid d) linolenic acid

3- Essential Fatty acid that should be in human diet include one of the following:

- a) Palmietic acid b) stearic acid c) oleic acid d) arachidonic acid

4- A non essential Fatty acid include which of the following:

- a) linoleic acid b) linolenic acid c) palmitic acid d) arachidonic acid

5- A non essential Fatty acid include which of the following:

- a) linoleic acid b) linolenic acid c) stearic acid d) arachidonic acid

6- Which is a major constituent of lung surfactant

- a) lecithin b) cholesterol c) glycerol d) Phosphaditic acid

7- Which is a major constituent of myelin sheath

- a) lecithin b) cholesterol c) cephalin d) Phosphaditic acid

8- The precursor compound of phospholipids is

- a) lecithin b) cholesterol c) glycerol d) Phosphaditic acid

9- The precursor compound of prostaglandins is

- a) lecithin b) cholesterol c) glycerol d) arachidonic acid

10- An unsaturated fatty acid with 3(three) double bonds is

- a) linoleic acid b) linolenic acid c) palmitic acid d) arachidonic acid

11- An unsaturated fatty acid with 2(two) double bonds is

- a) **linoleic acid** b) linolenic acid c) palmitic acid d) arachidonic acid

12- An unsaturated fatty acid with 4(four) double bonds is

- a) linoleic acid b) linolenic acid c) palmitic acid d) **arachidonic acid**

13- The number of double bonds in linolenic acid:

- a)one b) two c) **three** d) four

14- The number of double bonds in Arachidonic acid:

- a)one b) two c) three d) **four**

15- The number of double bonds in linoleic acid:

- a)one b) **two** c) three d) four

16- Fatty acid linked to glycerol by.....

- a) **Ester bond** b) glycosidic bond c) peptide bond

.....

Enumerate functions for the following:

A) Functions of lipids:

- (1) An efficient source of energy (1 gm of fat about 9 cal).
- (2) It serves as thermal insulators in the subcutaneous tissues and around certain organs e.g kidney
- (3) Lipids act as electrical insulators, allowing rapid propagation of the impulses along the myelinated nerves.
- (4) Source for Fat-soluble vitamins and essential fatty acids.
- (6) Cholesterol and its important derivatives as bile acids and steroid hormones.

B) Functions of Cholesterol:

1- cholesterol is a major constituent of plasma membrane.

2-synthesis of Bile acids: There are 2 types: cholic acid and chenodeoxycholic acid.

3- synthesis of Steroid hormones:

(A) Sex hormones: estrogens, androgens and progesterone.

(B) cortisone and aldosterone.

3- synthesis of Vitamin D : synthesized in the skin from cholesterol and activated in liver and kidney.

C) Importance of essential fatty acids:

1- Formation of phospholipids and cholesterol ester.

2- Enter in structure of cell membrane and are required for the fluidity of cell membrane.

3- They protect against atherosclerosis and coronary heart disease.

4- Arachidonic acid is a precursor of eicosanoids (e.g. prostaglandins).

Define the following:

Rancidity: is the development of bad odor and taste in the fat

Match from column A to the suitable answer in column B

A	B
1- <u>Rancidity</u> (d)	a- synthesized from cholesterol
2- Arachidonic acid (c)	b- An unsaturated fatty acid with 3 double bonds
3- Steroid hormones (a)	c- a precursor of prostaglandins
4- linolenic acid (b)	d- development of bad odor and taste in the fat
5- Glycogen (e)	e- a storage form of carbohydrates in liver and muscles

.....

Questions on Lecture of protein biochemistry

Choose the correct answer:

1- The metabolizable forms of amino acids are....

- a- L-aminoacids
- b- D-aminoacids
- c- R-aminoacids
- d- None of these

2- Protein denaturation leads to all of the following effects EXCEPT:

- a- Loss of catalytic activity of enzymes
- b- Loss of biological functions of hormones
- c- Decreased solubility
- d- **Decreased digestibility**

3- All the following proteins are of high biological value EXCEPT

- a- Albumin
- b- **Histones**
- c- Glutelins
- d- Globulins

4- All the followings are globular proteins EXCEPT

- a- Albumin
- b- Globulin
- c- **Insulin**
- d- **Collagen**

5- During denaturation of proteins all the bonds are destroyed except:

- a) The hydrogen bonds
- b) The ionic bonds
- c) **The peptide bonds**

6- High biological value proteins are:

- a) **Albumin and globulin**
- b) Protamines and glutelins
- c) Globulins and histones

a) Aspartate

b) Valine

c) Phenylalanine

7- Arginine is an example of

a) Acidic amino acids

b) Basic amino acids

c) Neutral amino acids

8- The only amino acid that is purely ketogenic is:

a) Lysine

b) Leucine

c) Isoleucine

9- Essential amino acids include all of the following effects EXCEPT

a) Aspartate

b) Tryptophan

c) Valine

d) Phenylalanine

10- Scleroproteins include:

a) collagen, elastin

b) collagen, albumin

c) albumin, globulins

11- Acidic amino acids are:

a- mono-amino, monocarboxylic

b- mono-amino, dicarboxylic

c- di-amino, monocarboxylic

12- Which one of the following nutritional disorders describes features of starvation in adults and marasmus in children?

a- very high energy daily intake

b- severe energy daily deficiency

c - Obesity

e - None of the above.

13-Which of the findings is present in child diagnosed as a case of kwashiorkor?

- a. Shows increased serum albumin.
- b. Appears plump due to increased adipose tissue.
- c. Displays abdominal and peripheral edema.
- d. Has markedly decreased weight for height.

14- An important etiological factor in kwashiorkor is:

- a. Excess dietary methionine
- b. Steatorrhoea
- c. Anaemia
- d. Dietary protein deficiency
- e. Dietary mineral deficiency

Define :-

1- **proteins:**

2- **Denaturation of proteins:** It is the destruction of the secondary, tertiary and quaternary structures of a protein molecule.

3- **High biological value proteins:** Contain all essential amino acids, and easily digested. E.g. albumin, globulins and glutelins.

4- **Essential amino acids:** Not synthesized in the body and must be supplied in the diet.

5- **Fibrousproteins:** have an axial ratio greater than 10 . The polypeptide chains coiled in spiral or helix and cross linked by hydrogen bonds like keratin, collagen and fibrin

Enumerate:-

2- Functions of proteins:

- **Nutritional:** each 1 gram protein gives about 4 Kcal.
- **Catalytic role:** All enzymes are proteins in nature.

- **Hormonal role:** Most of hormones and all cellular receptors are protein in nature.
- **Immunological role:** The antibodies (immunoglobulins) that play an important role in the body's defensive mechanisms are proteins in nature.
- **Structural role (e.g. collagen)**
- **Plasma proteins** are responsible for most effective **osmotic pressure** of the blood

3- Functions of albumin and globulins

- Both function as protein carrier for transport of different elements, vitamins & hormones.
- Albumin keeps blood osmotic pressure.
- Antibodies are γ -globulins.

4- Effects of denaturation on proteins.

- Loss of catalytic activity of enzymes
- Loss of biological functions of hormones
- Decreased solubility
- Increased viscosity
- Increased digestibility e.g. cooking meat & eggs by boiling

5- Applications of Denaturation on proteins.

- Denatured proteins, e.g., cooked meat are easily digested.
- Avoidance of denaturation is important for biological samples used for determination of enzymatic, hormonal or protein contents.
- Detection of albumin in urine by heat coagulation test is based on denaturation by heat.

6- Five of essential amino acids.

Isoleucine – Lysine- Tryptophan – Methionine - Leucine

7- Two semi-essential amino acids.

Histidine - Arginine

8- Metabolic classification subtype of amino acids.

- a) glucogenic amino acids
- b) ketogenic amino acids
- c) mixed amino acids

9- Five types of neutral amino acids.

- 1 **Aliphatic amino acids:** e.g., Glycine, Alanine.
- 2 **Hydroxy amino acids:** e.g., serine, threonine, tyrosine.
- 3 **Aromatic amino acids:** e.g., tyrosine, phenylalanine.
- 4 **Sulfur-containing amino acids:** e.g., Cysteine, homocysteine, cystine
- 5 **Heterocyclic amino acids:** e.g., Tryptophan, histidine

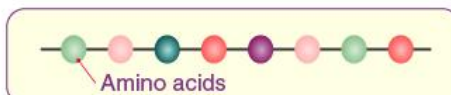
10- Four orders of structural organization of proteins.

- a. primary structure
- b. secondary structure
- c. tertiary structure
- d. quaternary structure

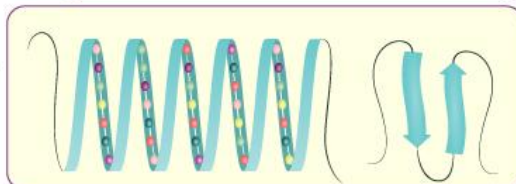
PROTEIN STRUCTURE



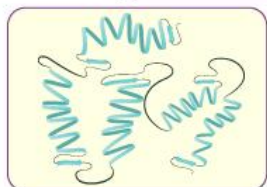
Primary Structure



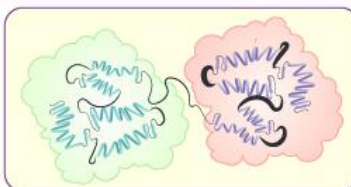
Secondary Structure
(α - Helix)



Tertiary Structure



Quaternary Structure



Compare between Marasmus and kwashiorkor diseases:

	Marasmus	Kwashiorkor
edema	Absent	present
Enlarged fatty liver	Absent	present
Changes in plasma proteins	Absent	present
Skin lesions	Absent	present

.....
Lecture: PH of Blood, Acidosis, Alkalosis, Buffer systems

1. At pH 7.4, the ratio of bicarbonate : dissolved CO₂ in plasma is

- a) 1 : 1
- b) 20 : 1
- c) 10 : 1
- d) 40 : 1

2. Quantitatively, the most significant buffer system in plasma is

- (A) Phosphate buffer system
- (B) Carbonic acid-bicarbonate buffer system
- (C) Lactic acid-lactate buffer system
- (D) Protein buffer system

3- The most significant buffer system in kidney is

- (A) Phosphate buffer system
- (B) Carbonic acid-bicarbonate buffer system
- (C) Lactic acid-lactate buffer system
- (D) Protein buffer system

4. A person was admitted in a coma (غيبوبة). Analysis of the arterial blood gave the following values: PCO₂ 16 mm Hg, HCO₃⁻ 5 mmol/l and pH 7.1. What is the underlying acid-base disorder ?

Normal Values: HCO₃ [22 - 28] - PH = 7.35 – 7.45 – P CO₂ [35 – 45]

a) **Metabolic Acidosis**

c) Metabolic Alkalosis

b) Respiratory Acidosis

d) Respiratory Alkalosis

5. A student is nervous for a big exam and is breathing rapidly, what do you expect out of the followings

a) Metabolic Acidosis

c) Respiratory Acidosis

b) Metabolic Alkalosis

d) Respiratory Alkalosis

6. Causes of metabolic alkalosis include all the following except.

a) Hypokalemia .

c) Recurrent vomiting

b) Thiazide diuretic therapy.

d) Mineralocorticoid deficiency.

7. Which of the following conditions will not cause respiratory alkalosis?

a) Fever

c) Laryngeal obstruction

b) Anxiety

d) Salicylate toxicity

9-A 45-year-old female with renal failure, missed her dialysis and was feeling sick, what could be the associated condition ?

a) Metabolic Acidosis

c) Respiratory Acidosis

b) Metabolic Alkalosis

d) Respiratory Alkalosis

10-The pH of the body fluids is stabilized by buffer systems. Which one of the following compounds is the most effective buffer system at physiological pH ?

a) Bicarbonate buffer

c) Protein buffer

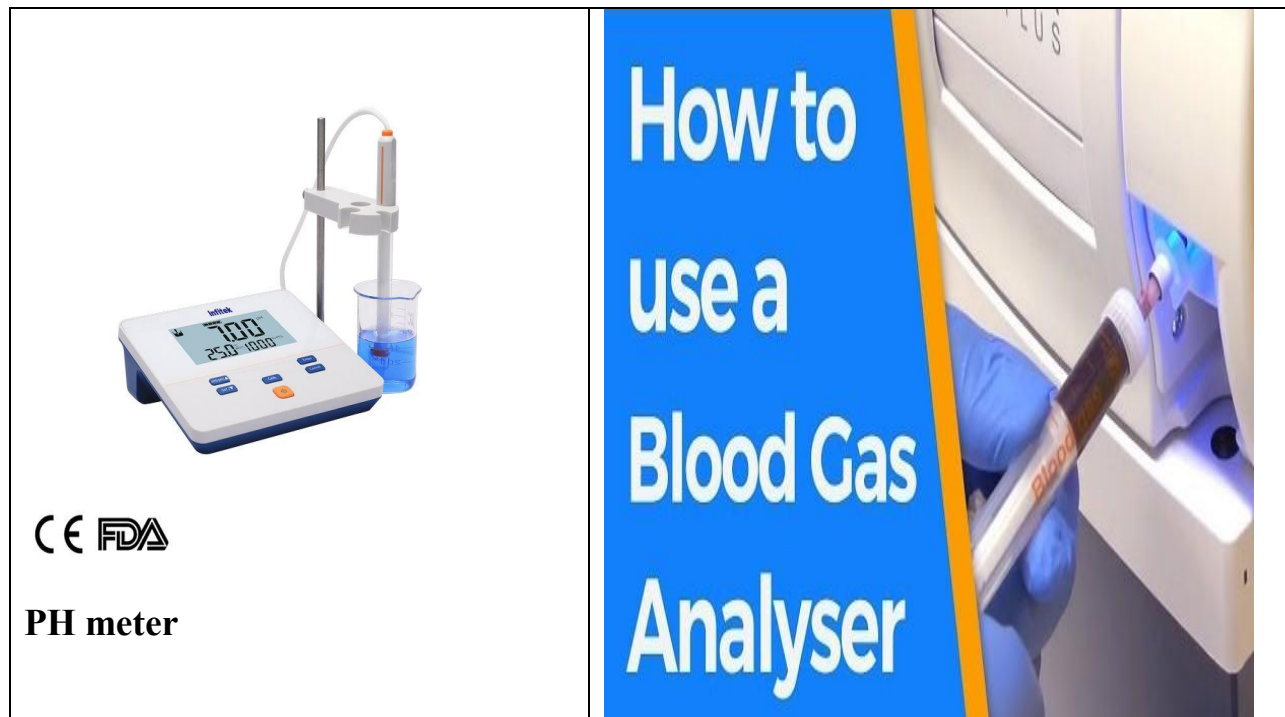
b) Phosphate buffer

d) None of the above

Complete:

1. PH measures:.....hydrogen concentration in the blood
2. The normal pH of the blood is7.35-7.45.....
3. Acidosis of blood means pH is.....below 7.35.....
4. Alkalosis of blood means pH is.....above 7.45.....
5. PH of blood could be measured by.....blood gas analyzer.....

6. PH of laboratory prepared solutions is measured bypH meter.....



True (✓) or False (X) and mention the right answer (Justify):

- a. Acidosis means: high pH is above 7.5 (x). Below 7.35
- b. Acidosis means: PH is below 7. 5 (x). Below 7.35
- c. Normal PH of arterial blood is between 7.35 – 7.45 (✓).
- d. Chronic obstructive lung disease (as Asthma) is complicated with acidosis (✓)..
- e. Chronic cigarette smoking usually leads to respiratory acidosis. (✓).

Lecture: Vitamins and minerals

I- Multiple Choice Questions:

1. Richest source of Vitamin D is

- (A) Cod liver oils (B) Margarine
(C) Egg yolk (D) Butter and Milk

2.The 7-dehydrocholesterol in skin is cleaved to form cholecalciferol by

- (A) Infrared light (B) Dim light
(C) Ultraviolet irradiation with sunlight. (D) electrical light

3. The most biologically potent Vitamin D metabolite is

- (A) 25-Hydroxycholecalciferol
(B) 1,25-Dihydroxycholecalciferol
(C) 24, 25-Dihydroxycholecalciferol
(D) 1,24 -hydroxycholecalceferol

4. Deficiency of vitamin D in adults causes

- (A) Anemia (C) Hypothyroidism
(B) Tuberculosis of bone (D) Osteomalacia

5.Which of the following vitamins is fat-soluble?

- a-Vitamin A b-Folic acid c- Biotin d- Niacin

6.Which of the followings is water-soluble vitamin?

- a) Vitamin A b) Vitamin D
c)Vitamin C d) Vitamin K

7.Which of the following statements describe water-soluble vitamins?

- a. Their absorption needs by bile salts
b. They need carrier protein

c. They can not be stored in the body

d. They include vitamin A&D

8. Which one of the following vitamins is derived from the carotenoid compounds?

a. Vitamin D

b. Vitamin B12

c. Vitamin C

d. Vitamin A

9.-All the following statements describe rickets Except:

a. It is caused by deficiency of vitamin D in infants

b. It is characterized by high plasma calcium and phosphorus levels

c. It is characterized by box shaped skull

d. There is knock-knee

10. -All the following vitamins belong to B complex vitamins except:

a. Riboflavin

b. Niacin

c. L. Ascorbic acid

d. Folic acid

11. FMN and FAD are derived from which one of the following vitamins?

a. Thiamin

b. Niacin

c. Riboflavin

d. Biotin

12. Which one of the following vitamins enters in the synthesis of coenzyme A?

a. Folic acid

b. Nicotinic acid

c. Pantothenic acid

d. Ascorbic acid

13. Pernicious anemia is caused by deficiency of one of the following vitamins;

a. Vitamin B6

b. Vitamin B1

c. Vitamin B12 (intrinsic factor)

d. Folic acid

N.B:- Pernicious anemia is a type of megaloblastic anemia in which the body isn't able to absorb vitamin B12 due to a lack of intrinsic factor in stomach secretions.

14.-Folate deficiency causes

(A) Microcytic anemia

(C) Iron deficiency anemia

(B) Hemolytic anemia

(D) Megaloblastic anemia

15.-The most important natural antioxidant is

(C) Vitamin D

(B) Vitamin E

(C) Vitamin B12

(D) Vitamin K

16.-Deficiency of vitamin D causes

(A) Rickets and osteomalacia

(B) Tuberculosis of bone

(C) Hypothyroidism

(D) Skin cancer

17.-Which one of the following statements describe vitamins?

a- They are required in large quantities for normal growth and health.

b- They cannot be synthesized by the body in adequate amounts and must be taken in diet.

c- They enter in the structure of tissues.

d- They are one of the major sources of energy to the body.

18.-All the following statements describe vitamin C EXCEPT:

a- It is derived from glucose

b- 90% of it, present in the form of L-ascorbic acid

c- it is involved in collagen synthesis

d- its deficiency causes beriberi

19.-Absorption of Vitamin B12 requires the presence of:

(A) Pepsin

(B) Renin

(C) Intrinsic factor

(D) amylase

20.-Deficiency of vitamin A causes

(A) Osteomalacia

(B) Night blindness

(C) Tuberculosis of bone

(D) Cheilosis

21.-Which of the following vitamin is stored in the liver?

- a. Thiamin
- b. VitaminC
- c. Pantothenic acid
- d. Vitamin A

22.-Which of the following organs is not a target organ for vitamin D action?

- A.Intestine B.Kidney C. Liver D. Bone

23.-Which form of vitamin K is produced by intestinal bacteria?

- a. Phylloquinone(K1)
- b. Menaquinone (K2)
- c. Phyllodione
- d. Menadione (K3)

24.-The synthetic form of vitamin k is:

- a. Phylloquinone(K1)
- b. Menaquinone (K2)
- c. Phyllodione
- d. Menadione (K3)

25.-Which vitamin is important for biosynthesis of clotting factors (II, VII, IX and X)?

- a.Thiamin b.Vitamin C c.Vitamin B12 d.Vitamin K

25- Prolonged prothrombin time (PT time, INR) occurs in case of vitamin.....deficiency.

- a.A b. E c.K d.b12

26- Loosening and falling of teeth, gums is swollen and infected and bleeds easily in vitamin deficiency.

- a.A b.C c.K d.b1

27- The coenzyme derived from vitamin B1 is.....

- a.PLP b.TPP c.NAD d.COASH

28- Biotin serves as a coenzyme forEnzymes.

- a.Dehydrogenases b.Lyases c. Carboxylases d.Ligases

29- About 70% of neural tube defect cases can be prevented by takingprophylactically during pregnancy.

- a. Vitamin D b. Vitamin E **c. Folic acid** d. Pantothenic acid

30- Vitamin D3 synthesis starts in skin fromas a precursor.

- a. Cholecalciferol
b. 7dehydrocholesterol
 c. Ergosterol
 d. Calcitriol

II- Match the following statements in column (A) with the suitable answer in the column (B)

A	The answer	B
1- It is considered as natural antioxidants	(d)	a. Vitamin D
2- Its deficiency causes rickets	(a)	b. Vitamin B12
3- Its coenzyme is FAD & FMN	(e)	c. Vitamin C
4- It needs intrinsic factor for its absorption	(b)	d. Vitamin E
5- It is given prophylactically in pregnancy to prevent spina bifida	(f)	e. Vitamin B2
6- Its deficiency causes scurvy	(c)	f. Folic acid

III- Put true or false and correct the false statement

- 1- Vitamins are organic nutrients which are required in small quantities for normal growth and health (√).
 2- The active form of vit D is 1 hydroxycholecalciferol (x).

1,25 di-hydroxycholecalciferol

3- Eye manifestations of vitamin B12 deficiency is night blindness (x).

vitamin A

4- Vitamin K2 is produced by intestinal bacteria (✓).

5- Deficiency of vitamin E produces scurvy (x). Vitamin C

6-Hypernatremia means an increase in potassium. (x). sodium

7-Hypokalemia means a decrease in potassium.. (✓).

IV- Complete:

1- Thiamine pyrophosphate is the coenzyme of vitamin B1..

2-Pellagra is characterized by: Diarrhia....., Demenia.....,
...Dermatitis.....

3- The coenzymes derived from niacin is ...FMN..... and ...FAD.....

4- Thiamin deficiency causes...Beri Beri.....

5- Ribofalvin is the scientific name of...B2.....

V-Put the suitable answer in front of the following statements:

1- The vitamin that has role in biosynthesis of clotting factors including (2,7,9,10) (Vitamin K).

2- The vitamin used in formation of COASH and Acyl carrier protein (Vitamin B5).

3-The vitamin that provide coenzyme PLP. (Vitamin B6 (Pyridoxine)

4- a glycoprotein secreted by the parietal cells of gastric mucosa and has role in B12 absorption. (Intrinsic factor)

5- Type of anemia due to folate deficiency (megaloblastic anemia)

VII-Short answered questions:

1- Compare between fat &water-soluble vitamins

Comparison between fat & water soluble vitamins

Item	Fat soluble vitamins	Water soluble vitamins
Source	Dietary fat	Water of cooked food
Absorption	Needs bile salts	Don't need bile salts
Carrier protein	Need carrier	Need no carrier
Storage	Can be stored in liver	Not stored in the body
Deficiency	Appear later	Appear early
Urine excretion	Can't be excreted	Easily excreted
Overdose	Leads to hypervitaminosis	Hypervitaminosis is rare

2- Enumerate three manifestations of vitamin A deficiency:

- Night blindness
- Lung infection
- Keratinization of skin

3- List three sources of vitamin K:

Sources:

- ❑ **Vitamin K1**(Phylloquinone): is natural vitamin K found in plants e.g. green leafy vegetables, cabbage, tomatoes and cereals.
- ❑ **Vitamin K2** (Menaquinone) : is produced by intestinal bacteria and found naturally in animal sources e.g. liver and egg yolk
- ❑ **Vitamin K3** (Menadione) : is synthetic vitamin.

Minerals

Sodium= Natrium (Hyponatremia نقص الصوديوم X Hypernatremia زيادة)

Potassium = Kalium(Hypokalemia نقص البوتاسيوم X Hyperkalemia زيادة)

I- Multiple Choice Questions-

1. All the following minerals are important constituents of bone and teeth except:

- a- Calcium
- b- Phosphorus
- c- Magnesium
- d- Sodium

2. Which one of the following nutritional disorders result from excessive fat accumulation?

- a- Undernutrition
- b- Malnutrition
- c- Obesity
- e- None of the above.

3. Which one of the following diseases result from excessive intake of high salt in Diet?

- a- Obesity
- b- Cancer of the breast
- b- Atherosclerosis
- d- Hypertension

4. - Excessive intake of CHO, fats and salt leads to development of all of the following except:

- a- Obesity
- b- Jaundice
- c. Atherosclerosis
- d- hypertension

5- All of the following are macro-minerals except:

- a-Sodium
- b- potassium
- c- magnesium
- d- zinc

6- All of the following are micro-minerals except:

- a-cobalt
- b- potassium
- c- iron
- d- zinc

Body calcium and phosphorus

1. What percentage of calcium in the human body is stored in bones and teeth?

- A. 50%
- B. 75%
- C. 90%
- D. 99%

☐ **Answer: D. 99%**

2. Which hormone primarily increases blood calcium levels?

- A. Insulin
- B. Parathyroid hormone
- C. Calcitonin
- D. Aldosterone

☐ **Answer: B. Parathyroid hormone**

3. Which of the following is a major function of calcium in the body?

- A. Oxygen transport
- B. Blood clotting
- C. Hormone synthesis
- D. DNA replication

☐ **Answer: B. Blood clotting**

4. Which vitamin is essential for calcium absorption in the intestine?

- A. Vitamin A
- B. Vitamin B12
- C. Vitamin C
- D. Vitamin D

☐ **Answer: D. Vitamin D**

5. Low levels of calcium in the blood (hypocalcemia) can lead to:

- A. Muscle relaxation
- B. Increased bone density
- C. Muscle spasms (tetany)
- D. Decreased nerve activity

☐ **Answer: C. Muscle spasms (tetany)**

6- Deficiency of calcium in adults cause:

- A. Hyperthyroidism
- B. Diabetes mellitus
- C. **Osteomalicia**
- D. Obesity

.....

2. Approximately what percentage of body phosphorus is found in bones and teeth?

- A. 50%
- B. 70%
- C. 85%
- D. 99%

☐ **Answer: C. 85%**

3. Phosphorus is a key component of a molecule responsible for energy transfer?

- A. DNA
- B. Hemoglobin
- C. ATP
- D. Insulin

☐ **Answer: C. ATP**

4. Which hormone helps regulate phosphate levels by increasing its excretion in the kidneys?

- A. Insulin
- B. Parathyroid hormone
- C. Aldosterone
- D. Calcitonin

☐ **Answer: B. Parathyroid hormone**

5. Which of the following conditions is associated with low phosphate levels?

- A. Hyperphosphatemia
- B. Hypophosphatemia
- C. Hypercalcemia
- D. Hyponatremia

☐ **Answer: B. Hypophosphatemia**

.....
Body sodium and potassium

1. What is the normal range of sodium in blood plasma?

- A. 120–130 mmol/L
- B. 135–150 mmol/L
- C. 150–160 mmol/L
- D. 160–170 mmol/L

☐ **Answer: B. 135–150 mmol/L**

2. Which hormone plays the most important role in sodium reabsorption in the kidneys?

- A. Insulin
- B. Aldosterone
- C. Parathyroid hormone
- D. Calcitonin

☐ **Answer: B. Aldosterone**

3. Sodium is primarily an important cation in which fluid compartment?

- A. Intracellular fluid
- B. Extracellular fluid
- C. Bone matrix
- D. Mitochondria

☐ **Answer: B. Extracellular fluid**

4. Which condition is caused by low sodium levels in the blood?

- A. Hyperkalemia
- B. Hypernatremia
- C. Hyponatremia
- D. Hypocalcemia

☐ **Answer: C. Hyponatremia**

5. A major function of sodium in the body is:

- A. Formation of hemoglobin
- B. Maintenance of osmotic pressure
- C. Storage of energy
- D. Production of enzymes

☐ **Answer: B. Maintenance of osmotic pressure**

.....

1. What is the normal range of potassium in blood plasma?

- A. 2.0–3.0 mmol/L
- B. 3.5–5.0 mmol/L
- C. 5.5–7.0 mmol/L
- D. 7.0–9.0 mmol/L

☐ **Answer: B. 3.5–5.0 mmol/L**

2. Potassium is mainly found in which compartment of the body?

- A. Extracellular fluid
- B. Blood plasma
- C. Intracellular fluid
- D. Bone

☐ **Answer: C. Intracellular fluid**

3. Which of the following is a key function of potassium?

- A. Blood clotting
- B. Nerve impulse transmission
- C. Oxygen transport
- D. Bone mineralization

☐ **Answer: B. Nerve impulse transmission**

4. Which condition refers to high potassium levels in the blood?

- A. Hypokalemia
- B. Hypernatremia

- C. Hyperkalemia
- D. Hyponatremia

☐ **Answer: C. Hyperkalemia**

5. Which organ plays the most important role in regulating potassium balance?

- A. Liver
- B. Heart
- C. Kidneys
- D. Lungs

☐ **Answer: C. Kidneys**

.....

1. What is the normal range of chloride in blood plasma?

- A. 70–85 mmol/L
- B. 90–100 mmol/L
- C. 96–106 mmol/L
- D. 110–125 mmol/L

☐ **Answer: C. 96–106 mmol/L**

2. Chloride is mainly found in which fluid compartment?

- A. Intracellular fluid
- B. Extracellular fluid
- C. Bone
- D. Muscle cells

☐ **Answer: B. Extracellular fluid**

3. Chloride plays an important role in:

- A. Oxygen transport
- B. Acid-base balance
- C. Hormone production
- D. Protein synthesis

☐ **Answer: B. Acid-base balance**

4. Chloride is a major component of which digestive substance?

- A. Bile
- B. Saliva
- C. Gastric acid (HCl)
- D. Pancreatic juice

☐ **Answer: C. Gastric acid (HCl)**

5. Low levels of chloride in the blood are referred to as:

- A. Hyperchloremia
- B. Hypochloremia
- C. Hyponatremia
- D. Hypokalemia

☐ **Answer: B. Hypochloremia**

II- Complete statements:

1. Main intracellular cation is.....**K**.....
2. Main extracellular cation is.....**Na**.....

3. Deficiency of potassium is called ...hypokalemia.....
4. Deficiency of sodium is calledhyponatremia.....
5. Body Mass Index (BMI) means...relative body weight to height.....
6. Waist to hip ratio normal in male equals..... ≤ 0.95
7. Normal random blood sugar should be less than ...140 mg/dl.....

.....

Lecture of Obesity

1. Obesity is defined as having a Body Mass Index (BMI) of:

- A. $\geq 23 \text{ kg/m}^2$
- B. $\geq 25 \text{ kg/m}^2$
- C. $\geq 30 \text{ kg/m}^2$
- D. $\geq 35 \text{ kg/m}^2$

☐ **Answer: C. $\geq 30 \text{ kg/m}^2$**

2. Which of the following is the most common cause of obesity?

- A. Genetic mutation
- B. Hormonal imbalance
- C. Excess caloric intake and low physical activity
- D. Infection

☐ **Answer: C. Excess caloric intake and low physical activity**

3. Which hormone is known as the “satiety hormone” that decreases appetite?

- A. Ghrelin
- B. Leptin
- C. Insulin
- D. Cortisol

☐ **Answer: B. Leptin**

4. Which hormone stimulates hunger?

- A. Leptin
- B. Insulin
- C. Ghrelin
- D. Thyroxine

☐ **Answer: C. Ghrelin**

5. Which of the following conditions is commonly associated with obesity?

- A. Hypoglycemia
- B. Osteoporosis
- C. Type 2 diabetes mellitus
- D. Anemia

☐ **Answer: C. Type 2 diabetes mellitus**

6. Central (abdominal) obesity increases the risk of:

- A. Asthma only
- B. Cardiovascular disease
- C. Skin infections only
- D. Eye disorders

☐ **Answer: B. Cardiovascular disease**

7. Which measurement is most useful to assess central obesity?

- A. Height
- B. Weight

- C. Waist circumference
- D. Arm span

☐ **Answer: C. Waist circumference**

8. Which of the following is a long-term complication of obesity?

- A. Increased immunity
- B. Hypertension
- C. Decreased cholesterol
- D. Improved insulin sensitivity

☐ **Answer: B. Hypertension**

9. Which lifestyle change is most effective for managing obesity?

- A. Increased sugar intake
- B. Sedentary lifestyle
- C. Balanced diet and regular exercise
- D. Skipping meals

☐ **Answer: C. Balanced diet and regular exercise**

10. Obesity increases the risk of which type of joint problem?

- A. Rheumatoid arthritis
- B. Osteoarthritis
- C. Gout only
- D. Bone fractures only

☐ **Answer: B. Osteoarthritis**

11- A person with body height of 180 cm and body weight of 100 kg, what is his BMI?

- A. 23 kg/m²
- B. 27.7 kg/m²
- C. 30.8 kg/m²
- D. 35 kg/m²

☐ **Answer C: 30.8 kg/m²**

12- A person with a BMI of 26 kg/m² is considered:

- A. Overweight
- B. Obese
- C. normal weight
- D. underweight

☐ **Answer: A. Overweight**

13- A person with a BMI of 17 kg/m² is considered:

- A. Overweight
- B. Obese
- C. normal weight
- D. underweight

☐ **Answer: D. Underweight**

.....

Lecture of Diabetes mellitus

1. What is Diabetes Mellitus?

- A. A disorder of protein metabolism
- B. A disorder of carbohydrate metabolism characterized by hyperglycemia
- C. A vitamin deficiency disease
- D. A bacterial infection

Answer: B

2. Which hormone is deficient in diabetes mellitus?

- A. Glucagon
- B. Insulin
- C. Cortisol
- D. Thyroxine

Answer: B

3. Type 1 diabetes is mainly caused by:

- A. Obesity
- B. Autoimmune destruction of beta cells
- C. Insulin resistance
- D. Excess sugar intake

Answer: B

4. Type 2 diabetes is primarily characterized by:

- A. Absolute insulin deficiency
- B. Insulin resistance
- C. Viral infection
- D. Increased insulin sensitivity

Answer: B

5. Which of the following is a classic symptom of diabetes?

- A. Bradycardia
- B. Polyuria
- C. Constipation
- D. Hair loss

Answer: B

6. Excessive thirst in diabetes is called:

- A. Polyuria
- B. Polydipsia
- C. Polyphagia
- D. Dysuria

Answer: B

7. Excessive hunger in diabetes is known as:

- A. Polydipsia
- B. Polyuria
- C. Polyphagia
- D. Anorexia

Answer: C

8. Which test is commonly used to diagnose diabetes?

- A. Hemoglobin test
- B. Fasting blood glucose test
- C. Urine protein test
- D. Liver function test

Answer: B

9. Normal fasting blood glucose level is:

- A. 50–70 mg/dL
- B. 70–110 mg/dL
- C. 120–150 mg/dL
- D. 160–200 mg/dL

Answer: B

10. A fasting blood glucose ≥ 126 mg/dL indicates:

- A. Hypoglycemia
- B. Normal condition
- C. Diabetes mellitus
- D. Dehydration

Answer: C

11. Which organ is primarily affected in diabetes?

- A. Liver
- B. Pancreas
- C. Kidney
- D. Heart

Answer: B

12. Which cells produce insulin?

- A. Alpha cells of langerhans
- B. Beta cells of langerhans
- C. Delta cells
- D. Acinar cells

Answer: B

13. Which complication is acute in diabetes?

- A. Neuropathy
- B. Retinopathy
- C. Diabetic ketoacidosis
- D. Nephropathy

Answer: C

14. Diabetic ketoacidosis is more common in:

- A. Type 2 diabetes
- B. Type 1 diabetes
- C. Gestational diabetes
- D. Prediabetes

Answer: B

15. Which of the following is a chronic complication?

- A. Hypoglycemia
- B. Ketoacidosis
- C. Retinopathy
- D. Hyperglycemia

Answer: C

16. Which drug is commonly used for Type 1 diabetes?

- A. Metformin
- B. Insulin
- C. Aspirin
- D. Antibiotics

Answer: B

17. Which drug is first-line for Type 2 diabetes?

- A. Insulin
- B. Metformin
- C. Paracetamol
- D. Steroids

Answer: B

18. Gestational diabetes occurs during:

- A. Childhood
- B. Adolescence
- C. Pregnancy
- D. Old age

Answer: C

19. Which lifestyle factor increases risk of Type 2 diabetes?

- A. Regular exercise
- B. Balanced diet
- C. Obesity
- D. Adequate sleep

Answer: C

20. Glycated hemoglobin (HbA1c) reflects:

- A. Blood sugar over 1 day
- B. Blood sugar over 2–3 months
- C. Blood pressure levels
- D. Kidney function

Answer: B